

Suresh Raghu

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EDUCATION

Indian Institute of Technology, Madras

B.Sc. in Programming & Data Science

Tamil Nadu, India

Oct. 2020 – Jun. 2026

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering

Madhya Pradesh, India

Oct. 2020 – Jun. 2024

PUBLICATIONS

Proper Scoring Rules for Agentic Uncertainty Quantification.

*Suresh Raghu**, *Satwik Pandey**, *Shashwat Pandey*

NeurIPS 2026

[arXiv:2605.24756](https://arxiv.org/abs/2605.24756) | Accepted (poster) at NeurIPS 2026 | * Equal contribution

- Developed **strictly proper scoring rules for uncertainty over LM-agent trajectories**, including a censored formulation for incomplete executions; showed that standard trajectory-level **ECE and Brier score** adaptations can reward miscalibrated forecasts, and demonstrated the resulting evaluation gap on long-horizon agent benchmarks with substantial trajectory censoring.

SELFDUBT: Uncertainty Quantification for Reasoning LLMs via the Hedge-to-Verify Ratio.

*Satwik Pandey**, *Suresh Raghu**, *Shashwat Pandey*

ICML 2026 Workshop

[arXiv:2604.06389](https://arxiv.org/abs/2604.06389) | Accepted (poster) at FAGEN Workshop @ ICML 2026 | * Equal contribution

- Proposed the **Hedge-to-Verify Ratio (HVR)**, a single-pass $\mathcal{O}(1)$ uncertainty signal that scores a reasoning trace by how much it hedges versus self-verifies. Traces with no hedging language are correct **96.1% of the time** (at 25.4% coverage), and fusing HVR with the model’s verbalized confidence beats sampling-based Semantic Entropy on discrimination ($p = 0.001$) at **10× lower inference cost**, across 7 models and 3 benchmarks.

Don’t Blink: Evidence Collapse during Multimodal Reasoning.

*Suresh Raghu**, *Satwik Pandey**

COLM 2026 Workshops

[arXiv:2604.04207](https://arxiv.org/abs/2604.04207) | Accepted (poster) at Sci-FM & Actionable Interpretability Workshops @ COLM 2026 | * Equal contribution

- Identified **“evidence collapse”**, a universal (9/9 model×dataset cells) decay of visual grounding during VLM reasoning that text-only entropy cannot detect, and designed a task-conditional **vision veto** that cuts selective risk by up to **1.9 pp** at 90% coverage on MathVista, HallusionBench, and MMMU.Pro.

RESEARCH & ENGINEERING EXPERIENCE

Looni Lab, Language Technologies Institute, Carnegie Mellon University

Remote

Independent Research Collaborator – Scientific Agents

July 2026 – Present

- **Agent Failure Analysis:** Audited long-horizon drug-design agents on **SMDD-Bench**; in 100 recoverable lead-optimization failures, **93% were primarily agent-control failures** and **89% had already generated a passing molecule or were one local edit away**, motivating a systematic study of how agent interfaces and state representations shape scientific-agent behavior.
- **Scientific Agent Harnesses:** Designed structured environments with immutable task specifications, canonical candidate state, evaluator-aligned verification, and task-bound **RDKit**, **ADMET-AI**, and **Boltz2** tools; improved **Qwen3.5-9B** lead-optimization success from **16.3% to 51.0%** on a matched 49-task evaluation.
- **Harness Engineering Study:** Led the manual harness-engineering investigation across SMDD-Bench and authored the accompanying technical article, *Is Human Taste Overrated in Harness Engineering?*; built reproducible **Harbor** / **Prime Env** infrastructure for budgeted scientific-agent rollouts, evaluator-backed rewards, trajectory logging, and distributed Boltz2 evaluation.

VFS Global

New Delhi, India

Senior Manager – AI (Founding Lead, AI Engineering)

May 2024 – Present

- **Adaptive Query Routing:** Architected the document-extraction pipeline around an adaptive router that sends deterministic documents to lightweight parsers and ambiguous ones to reasoning VLMs, reaching **99.1% field-level extraction accuracy**.
- **Uncertainty-Gated Extraction Cascade:** Used a cheap token-logit trigger to run **Semantic Entropy** only on suspect outputs, escalating flagged cases to a larger LLM and cutting critical extraction errors by **87%**.
- **Biometric Compliance Pipeline:** Proposed and built a two-stage visa photo-verification system consisting of a cheap image-quality gate followed by a VLM stage for pose and occlusion checks, reducing cases requiring manual intervention by **78%**.